

James B. Kane

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EDUCATION

Florida Atlantic University

Bachelor of Science in Electrical Engineering
Minor in Computer Science

Boca Raton, FL

Expected Graduation: May 2028

GPA: 3.78

SKILLS

Software: MATLAB, VHDL, Altium, Python, C, C++, C#, Git, SolidWorks

Hardware: Microcontrollers (ESP32, MSP430, Arduino, RP2040), Raspberry Pi, Multimeter, Oscilloscope, FPGAs (Artix-7), Soldering

EXPERIENCE

HID Global

Engineering Intern

Fort Lauderdale, FL

May 2026 – Aug 2026

- Automated the manual testing of PIV cards using a pivCLASS RPKCL40-P card reader by designing, constructing, soldering, and programming an electromechanical device, resulting in a reliable validation system capable of testing up to 120 card credentials per hour without physical human input.
- Synchronized 12 push-pull solenoids, a PIV smart card emulator, and a color sensor through custom C# and Python programs on a Raspberry Pi 3 & RP2040 (Microcontroller), leading to automated interactions with access readers that completed card tests in 30 seconds
- Designed and 3D-printed multiple SolidWorks part design iterations to solve alignment failures in a tightly-packed, vertically-staggered 12-solenoid array.

Florida Atlantic University

Learning Assistant (General Physics 1 & Calculus 2)

Boca Raton, FL

Aug 2025 – May 2026, Aug 2026 - Present

- Led student groups through complex physics and calculus problems using structured diagramming and equation application, directly improving student scores by an average of 30%
- Collaborated with faculty to implement active-learning strategies for a cohort of 80+ students.
- Served as a technical subject-matter resource, reinforcing core engineering concepts while mentoring freshman teams in collaborative problem-solving.

PROJECTS

L1-Class Rocket Flight Computer

In Development

- Developing a flight computer for a Level 1 High Power Rocket capable of recording altitude, velocity, acceleration, orientation, and position data locally and remotely using LoRa-based telemetry.

CLUBS & ORGANIZATIONS

Technology & Aerospace Club | Avionics Team Lead

Aug 2026 – Present

- Coordinating with fellow leadership to draft a preliminary design for an L3 rocket to compete in the 2027 International Rocket Engineering Competition (IREC).
- Selected barometers, inertial measurement units, and microcontrollers for an airbrakes system targeting 10k ft apogee, balancing measurement accuracy, sample rate, and power draw to meet control-loop and system integration requirements.

Owl's Racing (Formula SAE) | Electrical Subsystem Member

Jun 2025 – Aug 2026

- Assisted in the assembly and troubleshooting of the car's wiring harness.

IEEE-HKN | Recording Secretary

Apr 2026 – Present